

lab_code_16.c

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1 // Name: Md Shagor
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4 // Lab Report: 16
5 // Experiment Name: Transpose a Matrix
6
7
8 #include <stdio.h>
9
10
11
12 int main() {
13     int matrix[10][10], transpose[10][10];
14     int rows, cols, i, j;
15
16     // 1. Input Dimensions
17     printf("Enter rows and columns (max 10x10): ");
18     if (scanf("%d %d", &rows, &cols) != 2) {
19         printf("Invalid input.\n");
20         return 1;
21     }
22
23     // 2. Input Matrix Elements
24     printf("Enter matrix elements (%d x %d):\n", rows, cols);
25     for (i = 0; i < rows; i++) {
26         for (j = 0; j < cols; j++) {
27             printf("Element [%d][%d]: ", i, j);
28             scanf("%d", &matrix[i][j]);
29         }
30     }
31
32     // 3. Transpose Logic
33     // Original: matrix[i][j] -> Transpose: transpose[j][i]
34     for (i = 0; i < rows; i++) {
35         for (j = 0; j < cols; j++) {
36             transpose[j][i] = matrix[i][j];
37         }
38     }
39
40     // 4. Display Original Matrix (Good for comparison in Lab Reports)
41     printf("\nOriginal Matrix:\n");
42     for (i = 0; i < rows; i++) {
43         for (j = 0; j < cols; j++) {
44             printf("%d\t", matrix[i][j]);
45         }
46         printf("\n");
47     }
48 }
```

```
49 // 5. Display Transposed Matrix
50 // Note: The new dimensions are 'cols' rows and 'rows' columns
51 printf("\nTransposed Matrix:\n");
52 for (i = 0; i < cols; i++) {
53     for (j = 0; j < rows; j++) {
54         printf("%d\t", transpose[i][j]);
55     }
56     printf("\n");
57 }
58
59 return 0;
60 }
```